

TWENTY-FOUR CORE RESEARCH STACK

A Falsifiable Scientific Framework

Specification 24-CRS-1.0 · 26 September 2026

Status: operational standard. Not a theory of everything. Not an archaeological decoding.

This specification replaces the Duodecimal-Synthetic Framework (DSF-144) catalogue. It retains the arithmetic spine $12 \times \{1, 2, 4, 8, 12\} = \{12, 24, 48, 96, 144\}$ only as an index monoid and metrological codec. Scientific privilege is assigned exclusively to objects for which the integer 24 is forced by a theorem or a consistency condition. Every other numeral is a hyperparameter until it survives an adversarial control. The framework is true if and only if its stated predictions pass the protocols in Part IV. Failure of those protocols does not “weaken the poetry”; it retires the corresponding module.

1. Scope and non-claims

In scope: even unimodular lattices of rank 24; K3 topology and Mathieu/umbral/penumbral moonshine as typed data; critical bosonic transverse modes; weight-24 modular forms; SU(5) adjoint dimension; sexagesimal–duodecimal unit conversion; software that compiles these objects into codes, quantizers, and tests.

Out of scope, and hereby retired: derivation of physics from Sumerian tablets; replacement of binary logic by base-12 gates as a law of nature; 12-dimensional Lorentzian spacetime as a working theory; any 144-term $E_8 \times E_8$ “superforce”; any claim that neural width, SAT size, or transformer head count is forced by 12^2 .

2. Primitive objects

2.1 Index monoid

Let M be the multiplicative monoid generated by $\{2, 3\}$ inside the positive divisors of 144 and of 60. The distinguished chain is

$$12 \rightarrow 24 \rightarrow 48 \rightarrow 96 \rightarrow 144, \quad 12^2 = 144, \quad 24 \times 6 = 144, \quad 144 \times 5 / 12 = 60.$$

M explains why these integers recur in clocks, grids, and FFTs. M does not explain gauge groups, Calabi–Yau Hodge numbers, or model accuracy.

2.2 Forced 24-core

The core C_{24} is the tuple of objects for which 24 is not optional:

- (i) The set N_{24} of 24 even unimodular positive-definite lattices of rank 24 (Niemeier), including the Leech lattice Λ_{24} .
- (ii) Any K3 surface X , with $\chi(X) = c_2(X) = 24$, and the product identity $\chi(X \times X) = 576$.
- (iii) The 24 transverse modes of the critical bosonic string ($D - 2 = 24$).
- (iv) The discriminant modular form Δ of weight 24 and the ring of modular forms generated with E_4 , E_6 .

(v) $\dim \mathfrak{su}(5) = 24$.

(vi) Umbral moonshine: a typed assignment $X \mapsto (H^X, G^X)$ from Niemeier root systems to mock modular forms and finite groups, with Mathieu moonshine as the A_1^{24} case.

2.3 Historical residue

Mesopotamian area measure supplies $1 \text{ sar} = 12 \text{ cubits} \times 12 \text{ cubits} = 144 \text{ shekels of area}$, and sexagesimal place value with $60 = 12 \times 5$. These identities are implemented as a codec, not as cosmology.

3. Axioms

- A%d. Privilege. An integer n may be stored in the core namespace only if a named theorem, packing bound, representation identity, anomaly cancellation, or documented metrological ratio forces n .
- A%d. Typing. Every use of “24” must declare its type: lattice rank, Euler number, oscillator count, modular weight, group order factor, code length, or clock period. Untyped 24 is a hygiene violation.
- A%d. Non-derivation. No proposition of contemporary physics is deduced from the index monoid alone.
- A%d. Control spines. Any performance claim that cites 12, 24, 48, 96, or 144 as architecturally privileged must be repeated on the decimal spine $\{10, 20, 40, 80, 100\}$ and the hexadecimal spine $\{16, 32, 64, 128, 256\}$.
- A%d. Retraction. A module that fails its falsifier is removed from C_{24} . It may remain in a methods zoo under an unspined name.

4. Predictions

Each prediction is paired with a numeric or binary success criterion. Ambiguous “insight” is not a prediction.

- P%d. LLVQ dominance at fixed bit-width. On a frozen LLM checkpoint, 24-dimensional Leech-lattice vector quantization achieves strictly higher SQNR than E_8 product quantization and than scalar quantization at the same bits per coordinate, with codebook memory $O(1)$ in the first three shells.
- P%d. K3 checksum. Any pipeline that emits Euler characteristics for a declared product $K3 \times K3$ must return 576. Deviation is a software fault, not a new vacuum.
- P%d. Fibration classifier. A model trained on classified toric Calabi–Yau weight systems predicts presence of a K3 fibration on a held-out Kreuzer–Skarke subset at accuracy significantly above the majority-class baseline.
- P%d. Moonshine-conditioned vacua. Conditioning a flux-vacua generator on a K3-fibration bit plus a mock-shadow penalty increases the rate of tadpole-satisfying samples relative to an unconditioned twin trained on the same compute budget.
- P%d. Liftability. Claimed M_{24} (resp. M_{23}) symmetries of the odd Leech lattice VOA fail to lift. A certificate function returns false.
- P%d. Mukai bound. No symplectic automorphism group of a complex K3 larger than M_{23} is produced by any discrete-symmetry module in the stack.
- P%d. Anomaly lock. Any proposed field content whose worldsheet or gauge anomaly polynomial does not reproduce the known 24-related coefficient of a consistent theory is rejected by the monitor.

P%d. Codec exactness. Encode then decode of any sexagesimal place-value integer through the 12-spine recovers the input. 1 sar converts to 144 area-shekels exactly.

P%d. Hygiene. Unannotated occurrence of {12,24,48,96,144} as a network width is flagged by the static analyser.

P%d. Adversarial filter. If a width-n method on the 12-spine is statistically indistinguishable from its decimal and hexadecimal twins on a pre-registered metric, n is not promoted to C_24.

5. Falsifiers

A falsifier is a possible observation that obliges deletion or demotion. These are designed to be cheap to run.

ID	Module	Falsifier	Action on failure
F1	LLVQ	$SQNR \leq E_{8P}$ at equal bits on the registered checkpoint	Demote Leech quantizer; keep E_{8P}
F2	K3 checksum	Emitted $\chi(K3 \times K3) \neq 576$ after bug-free arithmetic	Halt pipeline; do not publish vacua
F3	Classifier	Held-out accuracy $\leq$ majority baseline + 2 s.e.	Do not claim K3 prediction as a primitive
F4	Vacuum prior	Conditioned hit rate $\leq$ unconditioned at equal budget	Drop moonshine regularizer
F5	VOA filter	Published lift of M_{24} through odd Leech VOA with complete proof	Invert the filter; update axiom set
F6	Mukai gate	Accepted example of symplectic Aut $> M_{23}$ on a K3	Relax the gate
F7	Anomaly monitor	Accepted consistent theory that the monitor rejects, or inconsistent theory it accepts	Patch the polynomial; do not waive
F8	Codec	Round-trip mismatch or sar $\neq$ 144 shekels	Fix arithmetic; historical claim paused
F9	Hygiene	Silent passage of unannotated spine widths	Linter is non-compliant
F10	Controls	12-spine wins on style but not on pre-registered metric	Item stays in zoo only

6. Operational stack

6.1 Functor F (the only closure)

F is the compiled chain Niemeier root system → umbral (or empty, for Leech) mock form and finite group → Construction A/B code → LLVQ kernel. Closure is the existence of this implementation plus the three arithmetic identities of §2.1. There is no further metaphysical map.

6.2 Repository layers

Layer	Contents	Admission rule
core/	N ₂₄ , Λ_{24} , G ₂₄ , Δ , K3 elliptic genus, su(5), D=24, codec	FORCED objects only
stack/	LLVQ, holy compiler, classifiers, generators, linters, monitors, digital twin	Implements a prediction in §4
zoo/	All former DSF items whose integer is a width	Unspined names; control-twin attached

6.3 Status of the original 144 items

FORCED (remain in core, renamed to their mathematical names): Niemeier family; $\chi(K3)=24$; Ramanujan τ / Δ ; bosonic 24 modes; $\dim SU(5)=24$; modular weight 24.

STRUCTURAL (stack or codec, not physics foundations): sexagesimal codec; 24-cell as a 4-polytope; dual-24 code maps; 48/96 geodesic and FFT lengths; $\chi(K3 \times K3)=576$.

ADMISSIBLE (zoo): approximately 110 width-prefixed methods (ResNets, PINNs, QKD channels, and the like).

MISLABELED or REJECT (erratum, do not ship under original titles): 12D heterotic spacetime; M₁₂ as Lorentzian 12-fold; 96-dimensional exceptional Jordan algebra; 96-dimensional Weyl group; 144-dimensional E₈; 144-term superforce; 144-cube spectrum; Sumerian convergence as a physical loop; SU(5) “48-particle field”; 96-loop QED as a calculation.

7. Protocols (minimum viable science)

Protocol A — Quantizer

Fix checkpoint, bit-width, evaluation split, and random seeds before training. Report SQNR, perplexity, and bits per coordinate against scalar, E₈P, and LLVQ. Memory of the codebook must be independent of 196560. Pre-register the comparison on a public issue tracker.

Protocol B — Geometry

Unit-test $\chi(K3)=24$ and $\chi(K3 \times K3)=576$. Train the fibration classifier with a frozen split of Kreuzer–Skarke weights. Publish calibration plots. Do not feed predicted Hodge numbers into phenomenology until the checksum passes.

Protocol C — Controls

For any zoo method advertised as “naturally 96” or “naturally 144,” clone the experiment at the nearest decimal and hexadecimal widths. Use a two-sided test on the pre-registered metric. Promote only on rejection of equivalence.

Protocol D — Symmetry budget

No discrete gauge or flavor claim enters stack/ without a character table and a declared target: M_{24} , a Niemeier Weyl group, or explicit NON-24. Mukai’s theorem caps symplectic K3 actions at M_{23} .

8. What would make the whole framework false

The stack as a scientific instrument fails if either (a) no module in core/ can be executed with a passing protocol, or (b) every performance advantage attributed to 24 is reproduced on the control spines. In that event the correct publication is not a revised poem of 144 items. It is a note that 24 remains a distinguished integer in lattice theory, K3 geometry, and critical strings, and that the index monoid is only a calendar and a grid.

The stack succeeds if Protocols A–D run, F1–F10 are live, and at least LLVQ or the K3 classifier plus checksum delivers a result that the control spines do not. That is a sufficient condition. It is also a necessary one.

9. Normative statement

24-CRS is a constraint on which integers may be treated as structure. It is not a unification of forces, not a base-12 replacement of computation, and not a reading of tablets. Sumer supplies a codec. Niemeier, K3, Δ , and $D - 2 = 24$ supply the core. Software supplies the tests. Everything else is a zoo.

Version 1.0 is frozen against narrative expansion. New items enter only by presenting a forcing theorem or a passed protocol. That rule is the framework.